

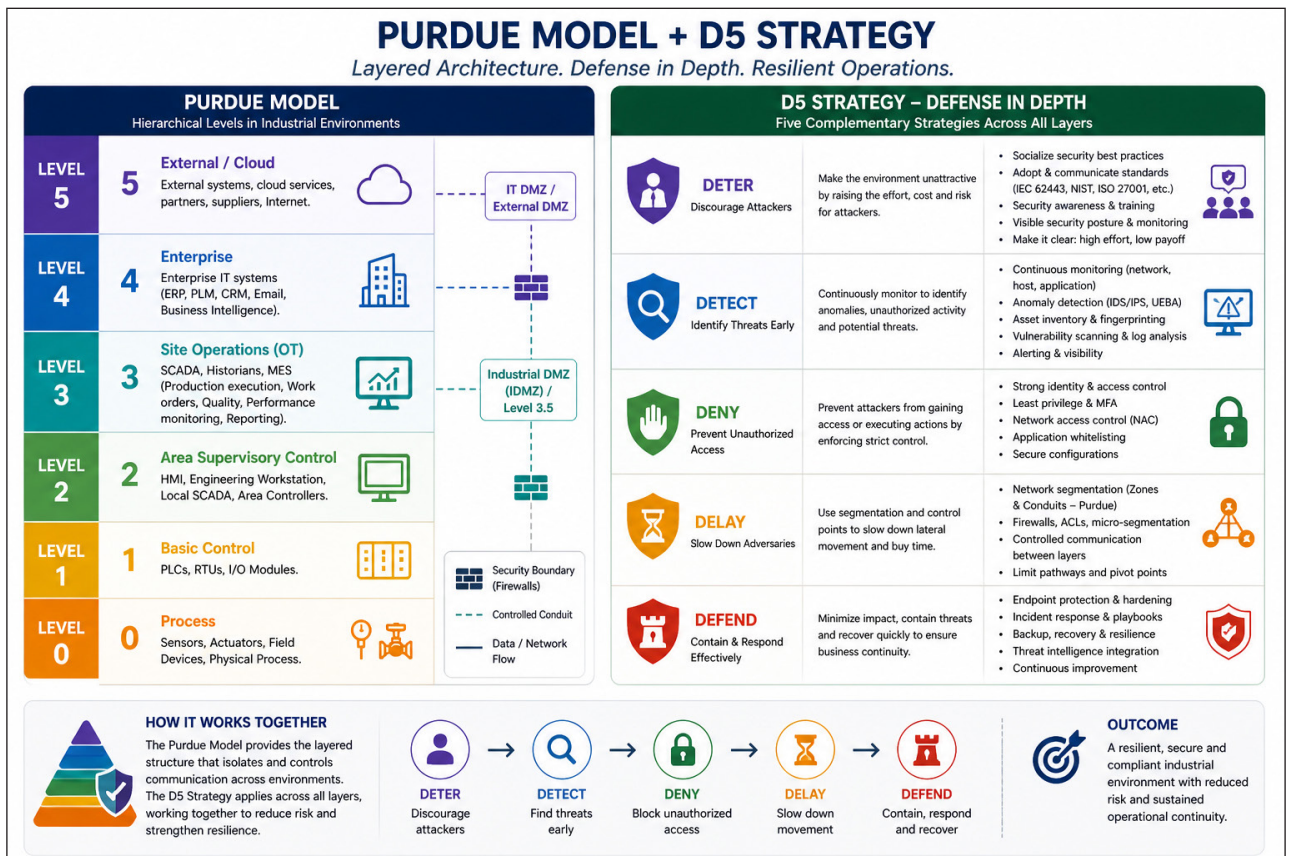


Figure 2: Purdue Model (Image source: AI generated)

account for increasing integration with Internet-based services. Data exchange at this level typically occurs through APIs, cloud integrations, or secure data transfer mechanisms.

Communication between Level 4 (enterprise IT) and Level 5 is governed by standard IT security controls such as firewalls, API gateways, and secure access mechanisms. These controls ensure that external connectivity does not introduce undue risk to enterprise or operational environments.

While the Purdue Model provides the architectural blueprint for our network, a static map is not enough to stop a modern adversary. To protect this hierarchy, we must layer our security controls through a defense-in-depth (D5) strategy, ensuring that if one layer of the Purdue Model is compromised, four more stand in the way of the physical process.

Defense-in-depth is a layered security strategy where multiple controls are applied across different levels of a system to isolate risks, limit the impact of threats, and provide redundancy in protection.

Deter: The primary goal is to discourage a potential adversary by making the ‘cost’ of an attack—in terms of time, effort, and risk of exposure—exceed the perceived value of the target. We achieve this by projecting a robust security posture. When we publicly or internally demonstrate our strict adherence to the Purdue Model, our use of encrypted communication, and our 24/7 monitoring, we signal to attackers that our environment is a ‘hard target’. This psychological barrier is designed to turn a volunteered attempt into a perceived failed mission before the first packet is ever sent.

Detect: The objective of this phase is to continuously monitor the environment to identify unauthorised communications, anomalous behavior, and potential security events. This is achieved through network and host-level monitoring, log analysis, and alerting mechanisms that can flag deviations from established baselines. A well-maintained asset inventory and device base lining (including firmware versions, configurations, and known vulnerabilities) support effective detection by enabling the identification of abnormal patterns and suspicious activity.

Deny: Beyond simply blocking unauthorised users, this phase ensures that only verified, cryptographically ‘known’ entities can communicate. This is achieved by implementing a Public Key Infrastructure (PKI). By requiring every device—from a